

Model Understanding: The goal of this report is to build a model to predict if a mortgage loan was approved or denied. Approval or denial will serve as the binary response variable (1 = denial and 0 = approval). Metrics including confusion matrix figures, ROC curves, Lift curves, measures of fit, and cumulative gains curves will be used to evaluate the effectiveness of the models. Ensemble modeling will also be explored in this report, which is done by combining several models to create a modeling average for increased prediction accuracy.

Data Understanding: Home Mortgage Disclosure Act (HMDA) data will be used for this report. The data set contained over 17,000 total records. Some records were excluded from the models because they were each missing 7 values. Additional records were excluded due to having quartile range outliers for the *AppIncome_num* variable. *ActionDescription* was excluded from analysis because it is a 100% correlated variable with only two values, one with a 100% Denied = 1 rate and the other with a 100% Denied = 0 rate. It is a text description of the *Denied* response variable, so exclusion avoids data leakage and overfitting. Variables including *LoanPurposeDescription* and *FFIECMedianFamilyIncome_num* were binned. For example, loans for the purposes of home improvement and refinancing had very similar Denied = 1 rates. The difference in the response variable for the purpose of the requested loan can be seen in Figure 5. This resulted in the creation of two bins for that specific variable. The entire data set was then split for validation in the following proportions: 60% training, 30% validation and 10% test. Because a validation column is necessary for the following process, *COUNTY_FIPS* and other variables were binned by decision tree partitions after the creation of the validation column. Each variable was split until the R-square of the training set remained constant. This allows the model to make better predictions while avoiding overfitting. All variables used for modeling are of the nominal data type due to binning or text values.

Analysis: After the data is cleaned and split into validation sets, different types of predictive models are created and compared for accuracy. The model types used for analysis in this report are k-nearest neighbor, boosted neural network, logistic regression, and boosted tree.

Confusion Matrix Figures: Figure 1 displays every confusion matrix figure for each model. Sensitivity and Specificity assess the model's ability to correctly predict classes. Sensitivity indicates the percentage of predicted Denied = 1 classes that were truly denied, and specificity shows the percentage of predicted Denied = 0 classes that were truly not denied. False negative and false positive rates do the opposite by indicating the percentage of incorrect Denied = 0 predictions and incorrect Denied = 1 predictions respectively. The overall error is calculated by dividing the sum of incorrect predictions by the total number of cases in the data set. The area under the curve (AUC) is also included in this metric for all models except k-nearest neighbor.

ROC Curve: The ROC curve is a useful metric to compare predictive models. Figure 2 displays the comparison of ROC curves for each model, excluding k-nearest neighbor. The

classification models in this report predict the probability that an observation belongs to the Denied = 1 class. A cutoff value of 0.5 is used to decide if the probability is high enough to be classified as Denied = 1. If the predicted probability of a case exceeds 0.5, like 0.51 for example, the observation will belong to the target class (Denied = 1). The straight line represents a default model where observations are classified randomly. Curves closer to the upper left corner with an AUC closer to 1 signify a model better at correctly predicting the Denied = 1 response variable. As seen in Figure 2, the boosted neural network had the highest AUC prior to the creation of the ensemble model. Logistic regression has the lowest AUC, indicating it is the least capable of classifying cases correctly. Figure 6 shows the cumulative gains curve, which measures the percentage of total positive targets (Denied = 1) "gained" in a specific percentage of the number of total cases. Better models are represented by curves closer to the top left of the graph similar to the ROC curve. All models gain around 80% of correct loan predictions in the first 50% of the data.

Lift Curve: Like the ROC curve, the lift curve is a method of comparison. It is another metric that demonstrates the advantage of using predictive modeling rather than random classification. Flipping a coin serves as random classification for this case, and the lift is calculated as the ratio of cumulative correct predictions over the cumulative expected by random classification. Figure 3 shows the lift curves of each model. A curve beginning well above 1 then falling steeply towards one at the end is considered a good model. Each model has a lift curve which demonstrates these qualities. They all appear to begin above 2.75 and fall quickly to 1.

Conclusion: The measures of fit for all validation levels of each model are shown in Figure 4. K-nearest neighbor isn't included in this and other metrics because it does not produce probabilities in JMP to make predictions. Instead, the model classifies predictions based on close, surrounding data points. Therefore, no statistics are populated in the measures of fit table, and no curves are calculated in figures 2, 3, and 6. The logistic regression has the highest sensitivity of 61%, indicating that it is the best at correctly predicting Denied = 1 cases. This model, however, has the lowest AUC and is also prone to overfitting. Neural networks are advantageous due to assessing complex relationships between predictor variables. The boosted neural network also has the lowest false negative rate (incorrectly predicted Denied = 0 cases), which is important given the goal of developing a model to predict loan approval. A false negative prediction represents approving a loan that was truly denied, and more loss generally occurs from accepting bad loans. With that, ensemble modeling is beneficial by combining several models to minimize limitations and increase prediction accuracy. Based on the metrics in this report, the modeling average is the best overall option for prediction because it has the highest AUC, second lowest false negative rate, and the same overall error rate as most other models. A different model, however, could be chosen depending on the business objective.

References: Module 4-7 Resources

Appendix

Figure 1: Confusion Matrix Figures

	Sensitivity	Specificity	F. Neg.	F. Pos.	Error	AUC
K-Nearest Neighbor	43%	89%	20%	39%	24%	N/A
Neural Network Boosted	58%	86%	16%	38%	22%	0.7893
Logistic Regression	61%	87%	17%	32%	22%	0.7759
Boosted Tree	50%	88%	18%	37%	22%	0.7856
Average Model	53%	87%	17%	37%	22%	0.7908

Figure 2: ROC Curves

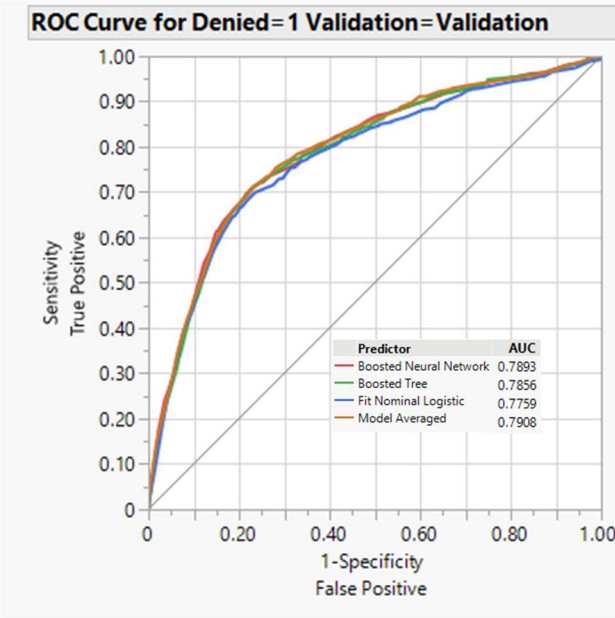


Figure 3: Lift Curves

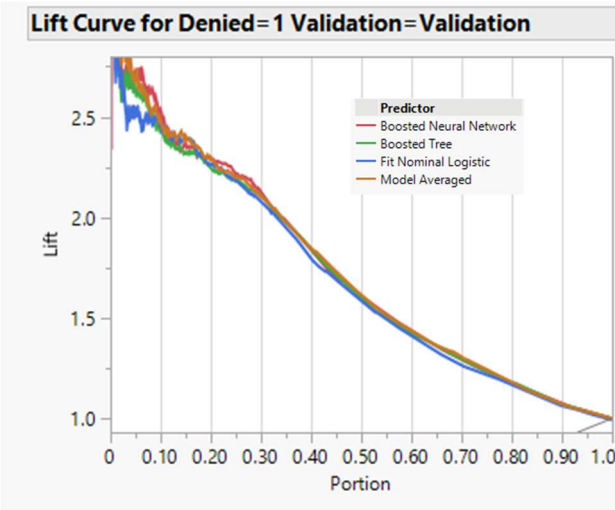


Figure 4: Measures of Fit/Model Comparison

Measures of Fit for Denied									
Validation	Creator	Entropy	Generalized	Mean Logp	RASE	Mean	Misclassification	Rate	N
Training	Neural Model NTanH(3)NBoost(3) - Save Formulas	2.468	0.2269	0.3391	0.4559	0.3827	0.2970	0.2084	10375
Training	Boosted Tree	0.2965	0.4140	0.4208	0.3659	0.2785	0.1987	10375	0.8439
Training	Fit Nominal Logistic	0.2534	0.3730	0.4403	0.3758	0.2818	0.2009	10375	0.8255
Training	Model Averaged	0.2591	0.3802	0.4369	0.3737	0.2872	0.1996	10375	0.8305
Validation	Neural Model NTanH(3)NBoost(3) - Save Formulas	0.1953	0.2981	0.479	0.3934	0.3082	0.2178	5148	0.7893
Validation	Boosted Tree	0.1814	0.2791	0.4873	0.3955	0.3019	0.2246	5148	0.7856
Validation	Fit Nominal Logistic	0.0517	0.0658	0.5645	0.3993	0.3021	0.2242	5148	0.7759
Validation	Model Averaged	0.1963	0.2995	0.4784	0.3929	0.3049	0.2214	5148	0.7892
Test	Neural Model NTanH(3)NBoost(3) - Save Formulas	0.1864	0.2870	0.4887	0.3988	0.3118	0.2262	1720	0.7892
Test	Boosted Tree	0.1926	0.2955	0.485	0.3974	0.3034	0.2262	1720	0.7936
Test	Fit Nominal Logistic	0.1671	0.2601	0.5003	0.4003	0.3016	0.2308	1720	0.7866
Test	Model Averaged	0.1994	0.3047	0.4809	0.3961	0.3062	0.2320	1720	0.7965

Figure 5: Loan Purpose Description vs Denied (before binning)

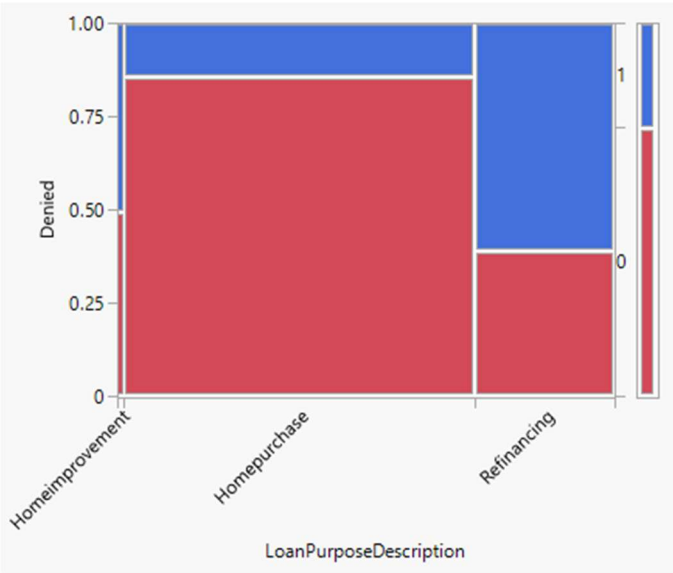


Figure 6: Cumulative Gains Curve

